

Vatsal Maisuria

Phone number: (+49) 1621630552 (Work) | **Email address:** vatsalmaisuria.77@gmail.com | **Website:** https://vatsal031177.github.io/vatsalmaisuria-portfolio/index.html | **LinkedIn:** https://www.linkedin.com/in/vatsal-maisuria/

ABOUT ME

Date of Birth: 03.11.2001 || **Gender:** Male || **Nationality:** India
Address: ZeppelinStrasse 134, 85051, Ingolstadt, Bayern, Germany
Work Permit: EU Blue Card Eligibility

WORK EXPERIENCE

FULL-STACK DEVELOPER – FREELANCER (PART - TIME) – 01/04/2024 – Current – INGOLSTADT, GERMANY

- Reduced latency for manual workflow processing by 45% by developing and deploying production ready LLM agents using Python, function calls, and structured output.
- Engineered human-in-the-loop (HITL) assessment pipelines and step-by-step tracking systems, successfully eliminating API hallucinations across critical business operations.
- Delivered a zero-to-one B2B SaaS workflow builder using React, Python, Flask, and GCP Cloud Run for dynamic task execution.
- Maintained 99.9% system uptime by integrating custom AI agents with PostgreSQL databases, CRM systems (HubSpot), and external REST APIs.
- Achieved 98% test validation coverage by containerizing client platforms with Docker and setting up automated GitHub/ GitLab release pipelines.

CONTENT MANAGEMENT SYSTEM DEVELOPER – CONTABO – 01/12/2025 – 30/04/2026 – INGOLSTADT, GERMANY

- Enforced Least Privilege Access models and integrated Multi-Factor Authentication across Content Management Systems, reducing lateral movement risks by 40%.
- Audited and mapped migrated data structures to guarantee strict compliance with GDPR, HIPAA, and PCI-DSS standards, successfully avoiding data spill incidents.
- Conducted continuous API penetration testing and configuration audits on the target Management Systems to remediate zero-day vulnerabilities before go-live.

Tech: PostgreSQL, Ubuntu VPS, RBAC, SOC Compliance, Bash, Linux, Docker, Prompt Engineering and Vibe Code

PROJECT MANAGER – ANNEXTECH SOFTWARE – 01/07/2022 – 31/03/2024 – GUJARAT, INDIA

- Led development of a Flutter and Spring Boot platform serving 10,000+ users with 99.9% uptime, 50,000+ daily API calls, and zero P1 incidents.
- Architected REST and gRPC microservices with JWT/OAuth2, OpenAPI 3.0, and circuit-breaker patterns, supporting three years in production.
- Built a multi-layer test suite achieving 85%+ coverage, preventing 60% of regressions before staging.
- Optimized GitHub Actions CI/CD with intelligent caching, delivering 40% faster builds and same-day production releases.
- Mentored three junior engineers through pair programming, increasing the team-wide first-pass rate to 78%.

SOFTWARE DEVELOPER – ANNEXTECH SOFTWARE – 01/07/2021 – 30/06/2022 – GUJARAT, INDIA

- Expanded product reach across iOS, Android, and desktop platforms by migrating legacy native Android applications (Java/C++) to a cross-platform Flutter architecture, increasing user adoption by 60%.
- Ensured 95% system uptime and reliable data flow by integrating front-end frameworks with backend cloud services, validating API responses through automated and manual testing with JUnit and Postman.

CERTIFICATIONS

CISCO, 21/08/2026
Cyber-Security

Link https://www.credly.com/badges/090b211c-fc84-4e8a-bae9-fc942164449b/linked_in?t=tk4pc7

Google Cloud, 22/08/2022
Build, train, and deploy ML models with Keras on google cloud

Link https://www.coursera.org/account/accomplishments/verify/2BWRHMFB3YYM?utm_source%3Dandroid%26utm_medium%3Dcertificate%26utm_content%3Dcert_image%26utm_campaign%3Dsharing_cta%26utm_product%3Dcourse

Google Cloud, 21/08/2022
Feature Engineering

Link https://www.coursera.org/account/accomplishments/verify/DX3JUZEAQTQ8?utm_source%3Dandroid%26utm_medium%3Dcertificate%26utm_content%3Dcert_image%26utm_campaign%3Dsharing_cta%26utm_product%3Dcourse

Coursera - Google Cloud, 30/07/2022
Google Cloud Big Data and Machine Learning

Coursera, 15/07/2022
CompTIA Security+ 701

Anthropic, 06/11/2026
Model Context Protocol: Advanced Topics

Write here the description

Link https://verify.skilljar.com/c/b43hufe8h953

Anthropic, 06/11/2026
Deploying Claude Enterprise with Confidence: The five decisions

Link https://verify.skilljar.com/c/frqkgmtp86ei

PROJECTS

01/06/2026 – 31/07/2026
ROS application for LiDAR Color Detection

- An end-to-end pipeline for fine-tuning an open-source LLM with QLoRA, evaluating it, merging/quantizing the adapter.
- Serving it in production behind a FastAPI endpoint, containerized with Docker and deployed on Kubernetes with autoscaling, monitoring, and CI/CD.

Link https://ai.studio/apps/1b5ade24-c32b-400a-a214-da4bb2fc32bf

01/05/2026 – 15/07/2026
Automation Flow Studio

AI workflow builder automating job-application tailoring, ATS scoring and more.
Tech: React, Python, Flask, Gemini API, Cloud Run

Link https://automation-flow-studio-42001982949.europe-west2.run.app

01/02/2026 – 30/04/2026
AI Resume Maker

To make resumes with the help of AI and Tailor it without credits and tokens.
Tech: TypeScript, Claude code, Gemini AI, Cloud Service, CSS TailWind, HTML5

Link https://flowcraft-resume-career-suite.ai.studio

01/10/2025 – 31/12/2025
Super Teams

- Created an application like Microsoft Teams with amazing features.
- Integrated tooling for Hosting and Joining with Auto-detect Language and can translate it in any language of the world.
- It works on live translation without any delay while having a Video chat everyone can use different languages at the same time.

Link https://github.com/Vatsal031177/teams-super

01/10/2025 – 31/12/2025

Multilingual-Conferencing Platform

- Created a scalable live collaboration application that supports multi-user video routing and secure meeting generation.
- Integrated OpenAI to process continuous audio streams and deliver real-time cross-captioning without noticeable latency (80ms).
- Future Scale: Overcoming global language barriers with real-time, zero-latency video conferencing.

Link https://github.com/Vatsal031177/teams-super

01/08/2025 – 01/10/2025

FullStack Postman Application (DevOps Project)

- Designed and implemented CI/CD pipelines showing hands-on experience with DevOps lifecycle.
- Integrated tooling for monitoring, logging and feedback loops showing you understand the deploy & monitor side of DevOps.
- Demonstrated collaboration & tooling chain integration linking source control, automated builds, tests, deployments, perhaps containerization and orchestration.

Link https://github.com/Vatsal031177/DevOps-Repository

01/06/2025 – 01/08/2025

Brand Analytics Site

- Sentiment analysis of social content.
- Growth prediction and optimal posting time prediction to maximise engagement.
- Real-time analytics and visualisations.

Link https://github.com/Vatsal031177/Personal-Brand-Analytics-Platform

01/05/2025 – 31/05/2025

Weather Update Application

- Integrated API calls to retrieve weather parameters, handled asynchronous logic, parsed JSON responses, and structured the UI to display temperature, humidity, wind speed and other key metrics.
- Designed a modular codebase and version-controlled it on GitHub.
- Ensured the app is responsive across device form-factors and designed for usability; implemented input validation, error handling and optional features such as unit conversion or dynamic UI feedback.

Link https://github.com/Vatsal031177/weather-application

01/04/2025 – 01/06/2025

Brute force Routing

- Assembling a structured repository for cyber security demonstrates passion for security, commitment to continuous learning.
- Available publicly to showcase your interest and capability in the cyber security domain.
- Consolidates knowledge on vulnerabilities, system hardening and defensive controls; leveraged Git/GitHub for version control and public sharing.

Link https://github.com/Vatsal031177/Cyber-Security-stuff-for-Knowledge-only

01/03/2025 – 31/05/2025

Automated ERP Inventory & Resource Management, TH Ingolstadt

Led development of a Flask-based ERP module for real-time resource tracking, capacity forecasting, and automated reporting. Integrated analytics dashboards to help project managers identify bottlenecks and optimise resource allocation across concurrent project teams.

01/03/2025 – 31/03/2025

Task Management Application

- Built a Task-Management Application in Dart/Flutter featuring task creation, editing, deletion, completion status, and local data persistence; structured UI with modular widgets and state management, enabling an intuitive and responsive mobile user experience.
- Architected the app with clear separation of UI, business logic and data layers, integrated local storage for persistent tasks, and maintained the codebase on GitHub to showcase clean version control and publicly accessible deliverables.

Link https://github.com/Vatsal031177/Task-Management-application-using-Dart

01/02/2025 – 01/04/2025

Autonomous Agentic AI

- Trained a voice-controlled agentic AI application that leverages TypeScript for robust backend and frontend logic orchestration.
- Performed comprehensive test scenarios and edge case test suites to rigorously evaluate system constraints under various acoustic conditions.
- Future Scope: Bridge between language and action with a highly tested, machine learning-based voice assistant.

Link https://github.com/Vatsal031177/Voice-Recognition-and-AI-agent/tree/main/Beast

01/01/2025 – 31/01/2025

WE-Tube Application

- Developed a YouTube-style clone application with video upload/playback, comments, likes/dislikes and search functionality, using JavaScript and modular front-end architecture.
- Built a responsive, interactive media platform clone implemented upload pipeline, video browsing UI, comment system and like/dislike feature demonstrating full-stack capability and user-centred design.

Link https://github.com/Vatsal031177/Youtube-Clone

01/11/2024 – 31/01/2025

Autonomous Navigation Agent, TH Ingolstadt

1. A reinforcement learning agent that perceives its environment and navigates autonomously toward defined goals in simulation.

Tech: Python, Reinforcement Learning, Reward Shaping, Simulation, TensorFlow, NVIDIA Jetson Nano

01/04/2024 – 11/07/2024

Heterogenous HPCs

- Research on OpenMP offloading, targeting CPU and GPU execution with minimal code duplication.
- Measured runtime modifications supporting heterogeneous workload execution and task migration.

01/02/2022 – 31/01/2023

Image Segmentation & Object Detection

- Trained and benchmarked YOLO/CNN segmentation models on custom-labelled datasets, deploying via TensorFlow Lite into a Flutter mobile UI for real-time on-device object detection for field technicians.

Tech: Python, YOLO, CNN, FCN, OpenCV, Flutter, Dart, Android SDK, Custom Dataset Labelling, Firebase Database

Link https://1drv.ms/w/c/78d26400eb5a16c2/IQBz8u6IzPmET73he7D88QgzASHBHp8z4c-euayJCNtXLAU?e=kFfSHQ

01/02/2022 – 31/01/2023

Image Segmentation and Object Detection

- Built a Flutter-based mobile app for real-time image segmentation and object detection using TensorFlow Lite.
- Integrated Firebase for secure user authentication, real-time database updates, and cloud storage of images and metadata.
- Optimized custom ML models for on-device inference, reducing processing time by 80%.

Link https://github.com/Vatsal031177/Object-detection-application-

01/02/2022 – 16/07/2022

Rocketry Development (I.S.R.O.)

- Led comparative evaluation of candidate structural materials for stiffness-to-weight and thermal resilience.
- Designed and executed scale-model test campaigns to validate performance trends and refine sizing assumptions.
- Synthesized results into technical reports and recommendations for subsystem design teams.

01/06/2021 – 10/11/2021

Captcha Code Development

- Designed a secure and lightweight CAPTCHA system without third-party dependencies.
- Implemented dynamic image rendering with obfuscation.
- Integrated session handling for CAPTCHA validation.
- Ensured compatibility with modern browsers and basic accessibility compliance

Link https://github.com/Vatsal031177/Vatsal031177

Autonomous Navigation Agent

Autonomous Agentic AI

Automated ERP Inventory & Resource Management

Brute force Routing

Multilingual-Conferencing Platform

Automation Flow Studio

ROS application for LiDAR Colour Detection

Fine-tuning LLM with QLoRA

Advance Dead Hand System

Autonomous Drone Cluster Network

01/06/2020 – CURRENT
Key Projects

Autonomous Drone Cluster Network | Advanced Dead Hand System | LiDAR Colour Detection Tools | LoRA-QLoRA Fine-Tuning | Image Segmentation and Object Detection | Automation Flow Studio | Multi-lingual Conference Platform | Brute Force Routing Tool | Agentic AI Voice System

PUBLICATIONS

2024
Heterogeneous High Performance Computing

- Authored peer-reviewed research examining performance trade-offs across multi-processor and GPU hardware combinations.
- Benchmarked compute throughput, memory bandwidth, and power efficiency, providing practical guidance for HPC system design decisions.
- Technologies: C++, CUDA, GPU Computing, MATLAB, Multi-Processor Architecture, Performance Benchmarking.

Publisher: Vatsal Kishorkumar Maisuria

2022
Image Segmentation and Object Detection

- Implemented pixel-wise image segmentation using clustering techniques, edge detection, and thresholding methods.
- Integrated an object detection pipeline capable of identifying multiple objects with bounding boxes and designed an FCN-based encoder-decoder model for semantic segmentation.
- Technologies: Python, YOLO, CNN, FCN, OpenCV, Flutter, Dart, Android SDK, Custom Dataset Labeling.

Publisher: Vatsal Kishorkumar Maisuria

EDUCATION AND TRAINING

17/03/2024 – 14/03/2026 Ingolstadt, Germany
MASTER OF ENGINEERING Technische Hochschule Ingolstadt

Field of study AI for Autonomous Systems (Incomplete)

01/07/2019 – 30/06/2023 Surat, India
BACHELOR OF ENGINEERING S. N. PATEL INSTITUTE OF TECHNOLOGY

Related Coursework: Object Oriented Programming, Data Structures, Software Engineering, Mobile Application Development, Design & Analysis of Algorithms, Theory of Computation, Compiler Design, Database Management System, Data Mining, Artificial Intelligence, Machine Learning, Distributed Systems, Computer Networks, Cyber Security, Cryptography and Network Security, IOT and Applications

Field of study Computer Science and Engineering | **Final grade** Overall Grade: 8.32/10 | **Thesis** Flutter Development (Grade: 10/10)

SKILLS

Programming Languages and Methodologies

Python | Dart | C++ | Bash | Java | SQL | TypeScript | Figma | SOC Compliance | Lean | Scrum | Grafana | Kanban | Agile

Databases and Cloud

Firebase | PostgreSQL | GCP | Docker | GitHub Actions | Kubernetes | Azure | Git | Prometheus | JIRA | Postman | Kali Linux | AWS | Linux | SQL Server | Firewall | Parrot Security OS | operating systems

Languages and Hobbies

English (Native - C2) | German (Intermediate - B1) | Gujarati (Native - C2) | Hindi (Native - C2) | Swimming | Chess | Football | Movies | Customize PCs

LANGUAGE SKILLS

Mother tongue(s): **GUJARATI (NATIVE - C2)** | **HINDI (NATIVE - C2)**

Other language(s): **ENGLISH (NATIVE - C2)** | **GERMAN (INTERMEDIATE - B1)**

HOBBIES AND INTERESTS

Music and Dancing

Playing Chess and Sudoku

Exploration

Watching Movies

HONOURS AND AWARDS

22/08/2022

Code Relay – S. N. Patel Institute of Technology and Research Centre

- Decreased system latency by 35% and cut deployment downtime by 60% by implementing robust security controls, optimizing resource management, and streamlining deployment pipelines.

21/08/2022

Feature Engineering – Google Cloud

- Describe Vertex AI Feature Store and compare the key required aspects of a good feature.
- Perform feature engineering using BigQuery ML, Keras, and TensorFlow.

Link https://www.coursera.org/account/accomplishments/verify/DX3JUZEAQTQ8?utm_source%3Dandroid%26utm_medium%3Dcertificate%26utm_content%3Dcert_image%26utm_campaign%3Dsharing_cta%26utm_product%3Dcourse

30/07/2022

Google Cloud Big Data and Machine Learning Fundamentals – Google Cloud

- Trained ML models with NumPy & scikit-learn, build & train supervised models for prediction & binary classification tasks (linear, logistic regression)
- Build a neural network with TensorFlow to perform multi-class classification, & build & use decision trees & tree ensemble methods

Link https://www.coursera.org/account/accomplishments/verify/MFWRUQFQ5WJF?utm_source%3Dandroid%26utm_medium%3Dcertificate%26utm_content%3Dcert_image%26utm_campaign%3Dsharing_cta%26utm_product%3Dcourse

RECOMMENDATIONS

Dr. Rutal S. Mahajan Head Of Department

Data Structures & Analysis of Algorithms, Algorithm Design & Implementation, Problem Solving & Logical Thinking, Self-learning & Continuous Improvement, Participation in Coding Competitions & Workshops

Email rutal.mahajan@snpitrc.ac.in | **Phone** (+91) 9426393096

Professor Bhavesh D. Patel Head of Department

Application UI Design, Front-end and Back-end Development, Figma & Dart, API Testing — Postman, JUnit, Mobile Application Development, Adaptability to New Technologies like IoT and Cybersecurity

Email bhavesh.patel@snpitrc.ac.in | **Phone** (+91) 9998471607